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Intensive Versus Conventional Blood Pressure Lowering After Successful Endovascular Thrombectomy: OPTIMAL-BP 1-Year Outcomes

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BACKGROUND: Intensive blood pressure (BP) lowering after successful reperfusion has resulted in short-term harm in acute ischemic stroke. However, it remains unclear whether these adverse effects persist over the long term.

METHODS: The OPTIMAL-BP (Outcome in Patients Treated With Intra-Arterial Thrombectomy–Optimal Blood Pressure Control) was a phase 3, multicenter, prospective, open-label, blinded end point, randomized controlled trial with 19 centers throughout South Korea. Patients who underwent endovascular thrombectomy for acute ischemic stroke caused by large vessel occlusion, achieved successful reperfusion of the occluded artery, and exhibited elevated BP (systolic BP ≥ 140 mm Hg) on 2 measurements obtained 2 minutes apart within 2 hours after recanalization were randomly assigned to receive intensive BP management (systolic BP target <140 mm Hg) or conventional management (systolic BP target, 140–180 mm Hg) for 24 hours after enrollment. This study was a 1-year follow-up extension of the OPTIMAL-BP. The primary outcomes were a modified Rankin Scale score of 0 to 2 at 1 year, indicating functional independence and all-cause mortality within 1 year. Adjusted odds ratios were estimated using multivariable logistic regression models adjusting for age, sex, onset-to-randomization time, and baseline National Institutes of Health Stroke Scale.

RESULTS: Among 306 randomized patients, 294 (96.1%) completed the 1-year follow-up. In the intention-to-treat analysis, functional independence at 1 year was numerically lower in the intensive BP management group than in the conventional group (40.5% versus 52.7%; adjusted odds ratios, 0.59 [95% CI, 0.34–1.00]; $P=0.051$). Consistent findings were observed in the per-protocol analysis (41.1% versus 54.7%; adjusted odds ratios, 0.56 [95% CI, 0.32–0.97]; $P=0.040$). One-year mortality and distribution of modified Rankin Scale changes from 3 months to 1 year did not differ between groups.

CONCLUSIONS: Intensive BP lowering targeting a systolic BP of <140 mm Hg resulted in worse functional outcomes at 1 year compared with conventional BP management. These randomized data suggest that early postthrombectomy BP management has durable effects on recovery and support current recommendations against intensive BP lowering.

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Key Words: blood pressure ■ ischemic stroke ■ odds ratio ■ reperfusion ■ thrombectomy

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Nonstandard Abbreviations and Acronyms	
BP	blood pressure
EVT	endovascular thrombectomy
mRS	modified Rankin Scale
OPTIMAL-BP	Outcome in Patients Treated With Intra-Arterial Thrombectomy–Optimal Blood Pressure Control
OR	odds ratio
SBP	systolic blood pressure

Optimizing blood pressure (BP) management after successful endovascular thrombectomy (EVT) has emerged as a potential strategy to improve functional outcomes in patients with acute ischemic stroke.^{1–3} Consequently, 6 randomized controlled trials have been conducted to evaluate the effects of intensive BP lowering compared with conventional BP management in the post-EVT setting.^{4–9} Of these, 2 trials reported results that intensive BP lowering failed to improve functional outcomes, while the remaining studies demonstrated neutral effects.^{5,6} Collectively, this evidence has reshaped clinical guidance: 2025 hypertension and 2026 early management in acute ischemic stroke guidelines recommend against lowering systolic BP (SBP) below 140 mmHg during the first 24 to 72 hours after successful recanalization, citing high-level evidence of potential harm.^{10–12}

However, an important unanswered question remains. Most post-EVT BP trials have relied on 3-month functional outcomes. Although widely accepted, this time point does not fully capture long-term recovery, adaptation, or decline after stroke.¹³ Functional trajectories may continue to evolve beyond 3 months, and it remains unclear whether the observed disadvantage associated with intensive BP lowering reflects a transient delay in recovery or a durable harm toward sustained disability. Therefore, it is important to evaluate whether differences in functional outcomes according to BP management strategies during the first 24 hours after successful recanalization persist into the long term (ie, at 1 year).

To address this gap in evidence, we present the 1-year outcome of participants in the OPTIMAL-BP (Outcome in Patients Treated With Intra-Arterial Thrombectomy–Optimal Blood Pressure Control) randomized controlled trial. We hypothesized that the effects of BP management strategies would persist at the 1-year mark.

METHODS

Data Availability

The anonymized OPTIMAL-BP data set will be available on reasonable request to the corresponding author.

Trial Design

The OPTIMAL-BP trial was a phase 3, multicenter, prospective, open-label, blinded end point, randomized controlled trial with 19 centers throughout South Korea. The trial protocol and consent forms were approved by the institutional review board of each participating hospital, and all patients or their legal representatives provided written informed consent. The study protocol is already published and available in the appendix.¹⁴ The trial was monitored by an independent data and safety monitoring board. This trial was registered at ClinicalTrials.gov and was completed. Consolidated Standards of Reporting Trials guidelines were used to report the findings.¹⁵

Participants

The trial design and procedures, including detailed inclusion and exclusion criteria, have been described elsewhere.¹⁴ Briefly, eligible participants were adults (age ≥20 years) who underwent EVT for acute ischemic stroke caused by large vessel occlusion, achieved successful reperfusion of the occluded artery, defined as a modified Treatment in Cerebral Infarction score ≥2b,¹⁶ and exhibited elevated BP (SBP ≥140 mmHg) on at least 2 measurements obtained 2 minutes apart within 2 hours after successful reperfusion. Key exclusion criteria included an SBP <140 mmHg following EVT, any contraindications to anti-hypertensive therapy, symptomatic intracerebral hemorrhage evident during or immediately after EVT, serious medical or surgical illness, and prestroke disability (modified Rankin Scale [mRS] score, 3–5). The decision to perform EVT in patients with acute ischemic stroke with large vessel occlusion was made according to the Korean Clinical Practice Guidelines.^{17,18}

Randomization and Masking

Participants were randomly assigned in a 1:1 ratio within 2 hours after reperfusion, with stratification by study center and baseline neurological severity (National Institutes of Health Stroke Scale score at admission, <15 points versus ≥15 points). Randomization was performed centrally using a secure, password-protected, web-based system with computerized random sequence generation. A permuted block design with a block size of 4 was applied. All follow-up examinations were conducted by investigators or study staff who were blinded to treatment allocation.

Interventions

Patients were randomly assigned to either an intensive BP management targeting an SBP of <140 mmHg or a conventional BP management targeting an SBP of 140 to 180 mmHg. All patients were treated in a stroke unit or an equivalent setting with continuous BP monitoring and guideline-based standard care.¹⁹ BP was measured serially at prespecified intervals using automated noninvasive monitoring devices applied to the non-paralyzed arm, or to the right arm in patients with coma or tetraparesis, according to the study protocol. Institutional protocols for intravenous BP-lowering therapy were in place, with nicardipine used as the first-line agent; alternative agents were permitted at the treating physician's discretion. In the conventional BP management group, vasopressors were not administered to maintain SBP targets when SBP fell below 140 mmHg; however, intravenous fluids and vasopressors could be used at the

physician's discretion to treat clinically significant hypotension. In both groups, the treatment goal was to achieve the assigned SBP target within 1 hour after randomization.

Outcomes

The primary efficacy outcome of the OPTIMAL-BP trial was a dichotomized analysis of the mRS score at 3 months (0–2, indicating functional independence, versus 3–6, indicating dependence or death). In this article, we report analyses of the 1-year outcomes of the trial. The primary outcomes were functional independence at 1 year, defined as an mRS score of 0 to 2, and all-cause mortality within 1 year. Secondary outcomes included the full distribution of mRS scores at 1 year, excellent functional outcome defined as an mRS score of 0 to 1 at 1 year, moderate functional outcome defined as an mRS score of 0 to 3 at 1 year, and a combined mRS score of 5 to 6 at 1 year. The 1-year mRS scores were determined in participants via telephone or in person by local certified medical staff who were blinded to the treatment allocation. The mRS score was determined by a structured interview using the Korean version of the mRS (<https://www.k-stroke-academy.co.kr>). Mortality data were obtained through linkage with the national death registry of Statistics Korea, which includes dates and causes of death through December 31, 2023, and was officially released in October 2024. Because the final patient was enrolled on November 29, 2022, all participants had at least 1 year of follow-up for mortality assessment.

Statistical Analysis

As with the analysis of the primary end point of the trial, the primary efficacy analyses of outcomes at 1 year were performed in the intention-to-treat population, with patients classified according to the group to which they were randomly assigned, regardless of the treatment received or protocol violations. The per-protocol analyses included patients who underwent the assigned treatment with no major protocol deviations. Binary logistic regression analyses were performed for the primary outcomes, and treatment effects were presented as odds ratios (ORs) with 95% CIs. In addition, risk differences with 95% CIs were calculated. Adjusted ORs were calculated using a multivariable logistic regression model adjusted for the same prespecified covariates used in the main analysis, including age, sex, onset to randomization time, and National Institutes of Health Stroke Scale score at admission. For the outcome of overall survival, we performed a Cox proportional-hazards model with the same adjustments as previous logistic regression models to estimate the adjusted hazard ratios with 95% CIs. The proportional hazards assumption was satisfied, as assessed by Schoenfeld residuals. For the secondary outcomes, the common OR representing a shift in the mRS scores at 1 year was estimated using ordinal logistic regression, with mRS scores of 5 and 6 merged for the shift analysis. Proportionality for this analysis was confirmed. Missing mRS values at 1 year were not imputed. The other secondary outcomes were analyzed using the same methods as those applied to the primary outcomes. Prespecified subgroup analyses were conducted for the primary outcome. Interaction terms between treatment assignment (intensive versus conventional management) and each subgroup variable were included in logistic regression models to evaluate whether the effect of treatment on functional

independence at 1 year varied across subgroups. The consistency of treatment effects among subgroups was assessed using these interaction models. Statistical analyses were conducted using R version 4.5.2 (R Foundation). All statistical tests were 2-sided, with $P < 0.05$ indicating statistical significance.

RESULTS

From June 18, 2020, to November 29, 2022, 306 participants were enrolled and randomized to the intensive management ($n=155$) and conventional management ($n=151$) groups. In the intention-to-treat analysis, there were 12 patients lost to 1-year follow-up because of immediate withdrawal of consent after randomization ($n=1$), transfer within 24 hours ($n=2$), loss to follow-up within 1 month ($n=1$), and loss to 1-year follow-up ($n=8$). In the per-protocol analysis ($n=280$), 2 patients assigned to receive conventional management crossed over and received intensive management. There were no crossovers among those assigned to receive intensive management. Protocol violations due to a prestroke mRS score ≥ 3 were identified in 7 patients in the intensive management group and in 5 patients in the conventional management group (Figure 1). Baseline demographics, clinical, and imaging characteristics were well balanced between groups (primary efficacy analysis in Table 1 and per-protocol analysis in Table S1). The mean age was 73.0 (SD, 11.6) years, and 41.2% were female. The mean National Institutes of Health Stroke Scale score at admission was 13 (SD, 6). Ninety-seven patients (33.0%) received an intravenous tissue-type plasminogen activator before EVT. All patients underwent EVT under conscious sedation or local anesthesia.

Primary Outcomes

In the intention-to-treat analysis, the proportion of patients who achieved functional independence at 1 year was statistically marginally lower in the intensive management group (40.5%) than in the conventional management group (52.7%), with a significant risk difference of -12.2% (95% CI, -23.5% to -0.9%) and an adjusted OR of 0.59 (95% CI, 0.34–1.00; $P=0.051$; Table 2; Figure 2A). In the per-protocol analysis, the proportion of patients who achieved functional independence at 1 year was significantly lower in the intensive management group (41.1%) than in the conventional management group (54.7%), with a significant risk difference of -13.5% (95% CI, -25.1% to -1.9%) and an adjusted OR of 0.56 (95% CI, 0.32–0.97; $P=0.040$; Table 3; Figure 2B). For 1-year mortality, 34 of 155 patients (21.9%) in the intensive management group and 27 of 151 patients (17.9%) in the conventional management group died in the intention-to-treat analysis. In the per-protocol analysis, deaths occurred in 32 of 149 patients (21.5%) in the intensive management group and

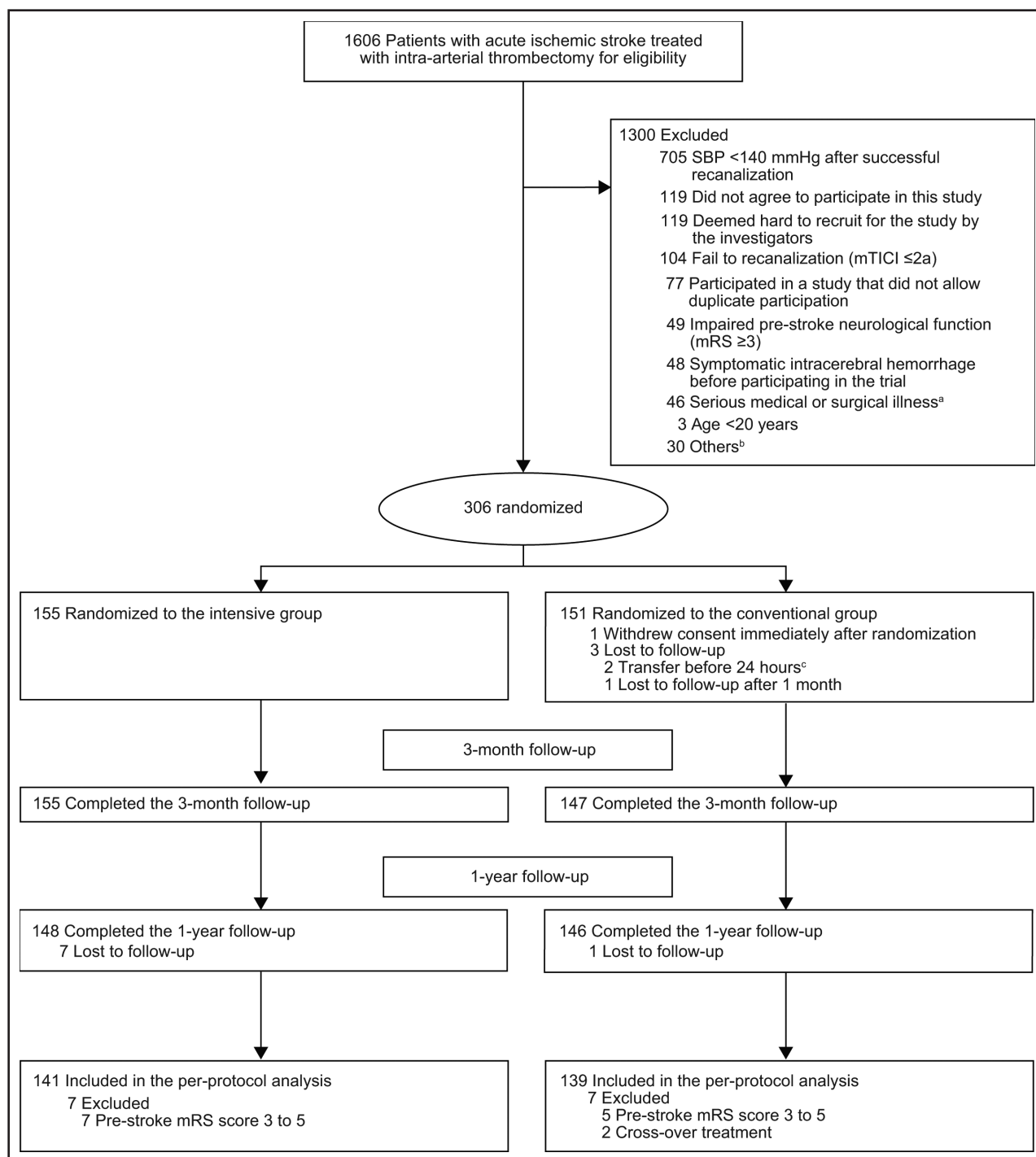


Figure 1. Participant flow diagram and 1-year follow-up loss.

^a Fifteen patients were excluded for terminal cancer, 14 with cardiac or aortic disease, 4 with severe anemia and hematologic disease, 4 with chronic kidney disease, 4 with pneumonia, 2 with sepsis, 1 with cholecystitis, 1 with hemothorax, and 1 with radius fracture. ^b Eleven patients were excluded for COVID-19 infection, 5 due to investigator error, 2 due to non-Korean citizenship, 2 due to immediate transfer to other hospitals, 2 due to no guardian, 2 for advanced dementia, and 6 for unknown reasons. mRS indicates modified Rankin Scale; mTICI, modified Thrombolysis in Cerebral Infarction; and SBP, systolic blood pressure.

24 of 146 patients (16.4%) in the conventional management group. No significant difference was observed between the intensive and conventional management groups in either the intention-to-treat analysis (risk difference, 4.5% [95% CI, -4.8% to 13.7%]; adjusted

OR, 1.31 [95% CI, 0.70–2.46]; $P=0.394$) or the per-protocol analysis (risk difference, 5.4% [95% CI, -3.8% to 14.7%]; adjusted OR, 1.36 [95% CI, 0.71–2.62]; $P=0.353$). Consistently, Kaplan-Meier curves showed no between-group difference in time-to-death within 1 year

Table 1. Demographic and Clinical Characteristics

Characteristics	Intensive management (n=148)	Conventional management (n=146)
Demographics		
Age, mean (SD), y	73.1 (12.2)	72.9 (10.9)
Sex, n (%)		
Female	62 (41.9)	59 (40.4)
Male	86 (58.1)	87 (59.6)
Medical history, n (%)		
Hypertension	115 (77.7)	109 (74.7)
Atrial fibrillation	73 (49.3)	69 (47.3)
Diabetes	64 (43.2)	61 (41.8)
Hyperlipidemia	59 (39.9)	54 (37.0)
Smoking	38 (25.7)	29 (19.9)
Previous stroke	35 (23.6)	29 (19.9)
CAOD	17 (11.5)	15 (10.3)
Active cancer	9 (6.1)	5 (3.4)
Congestive heart failure	6 (4.1)	7 (4.8)
PAOD	2 (1.4)	6 (4.1)
NIHSS score at admission, mean (SD)	13 (6)	13 (6)
0–5	11 (7.4)	15 (10.3)
6–15	83 (56.1)	81 (55.5)
≥16	54 (36.5)	50 (34.2)
Intravenous tissue-type plasminogen activator	43 (29.1)	54 (37.0)
Premorbid modified Rankin Scale, median (IQR)	0 (0–0)	0 (0–0)
Imaging findings		
Occlusion site		
MCA	100 (67.6)	99 (67.8)
ICA	29 (19.6)	29 (19.9)
VBA	15 (10.1)	12 (8.2)
MCA+ICA	3 (2.0)	2 (1.4)
ACA	1 (0.7)	3 (2.1)
PCA	0 (0.0)	1 (0.7)
ASPECTS, median (IQR)	9 (7–10)	9 (8–10)
TOAST classification, n (%)*		
Cardioembolism	73 (49.3)	75 (51.4)
Large artery atherosclerosis	39 (26.4)	43 (29.5)
Stroke of another determined cause	3 (2.0)	1 (0.7)
Undetermined negative evaluation	21 (14.2)	22 (15.1)
Undetermined 2 or more causes identified	12 (8.1)	5 (3.4)
Time parameters, median (IQR), min		
Onset to puncture time	379.0 (223.5–722.0)	357.0 (208.0–730.0)
Puncture to reperfusion time	30.5 (23.0–49.0)	31.0 (20.0–48.0)
Onset to reperfusion time†	416.0 (265.5–809.5)	408.5 (247.0–764.0)
Onset to randomization time	487.0 (327.5–840.0)	480.0 (310.0–820.0)

The table was presented as number (%), mean±(SD), or median (IQR). ACA indicates anterior cerebral artery; ASPECTS, Alberta Stroke Program Early Computed Tomography Score; CAOD, coronary artery obstructive disease; EVT, endovascular thrombectomy; ICA, internal carotid artery; IQR, interquartile range; MCA, middle cerebral artery; NIHSS, National Institutes of Health Stroke Scale; OPTIMAL-BP, Outcome in Patients Treated With Intra-Arterial Thrombectomy–Optimal Blood Pressure Control; PAOD, peripheral artery occlusive disease; PCA, posterior cerebral artery; TOAST, Trial of ORG 10172 in Acute Stroke Treatment; and VBA, vertebrobasilar artery.

*The TOAST classifications were determined based on a consensus of stroke neurologists in each participating hospital.

†The difference in the median onset to reperfusion time compared with the previously reported OPTIMAL-BP trial was attributable to the exclusion of patients without 1-year follow-up in the updated data set, resulting in a slight shift in the median value.

Table 2. Primary and Secondary Outcomes in the Intention-to-Treat Analysis

	Intensive management (n=148)	Conventional management (n=146)	Risk difference, % (95% CI)	Unadjusted OR (95% CI)	Adjusted OR* (95% CI)	P value
Intention-to-treat analysis						
Primary outcomes						
Functional independence at 1 y (mRS score of 0–2)	60/148 (40.5)	77/146 (52.7)	−12.2 (−23.5 to −0.9)	0.61 (0.38 to 0.97)	0.59 (0.34 to 1.00)	0.051
Mortality within 1 y	34/148 (23.0)	27/146 (18.5)	4.5 (−4.8 to 13.7)	1.31 (0.75 to 2.33)	1.31 (0.70 to 2.46)	0.394
Secondary outcomes						
mRS score increases at 1 y (shift analysis)	3 (1 to 5)	2 (1 to 5)		1.57 (1.04 to 2.37)	1.52 (0.99 to 2.35)	0.057
Excellent functional outcome at 1 y (mRS score of 0–1)	46/148 (31.1)	63/146 (43.2)	−12.1 (−23.0 to −1.1)	0.59 (0.37 to 0.96)	0.55 (0.31 to 0.97)	0.042
Moderate functional outcome at 1 y (mRS score of 0–3)	77/148 (52.0)	92/146 (63.0)	−11.0 (−22.2 to 0.2)	0.64 (0.40 to 1.01)	0.63 (0.37 to 1.08)	0.093
mRS score of 5–6 at 1 y	60/148 (40.5)	40/146 (27.4)	13.1 (2.4 to 23.9)	1.81 (1.11 to 2.96)	1.93 (1.10–3.45)	0.024

mRS indicates modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; and OR, odds ratio.

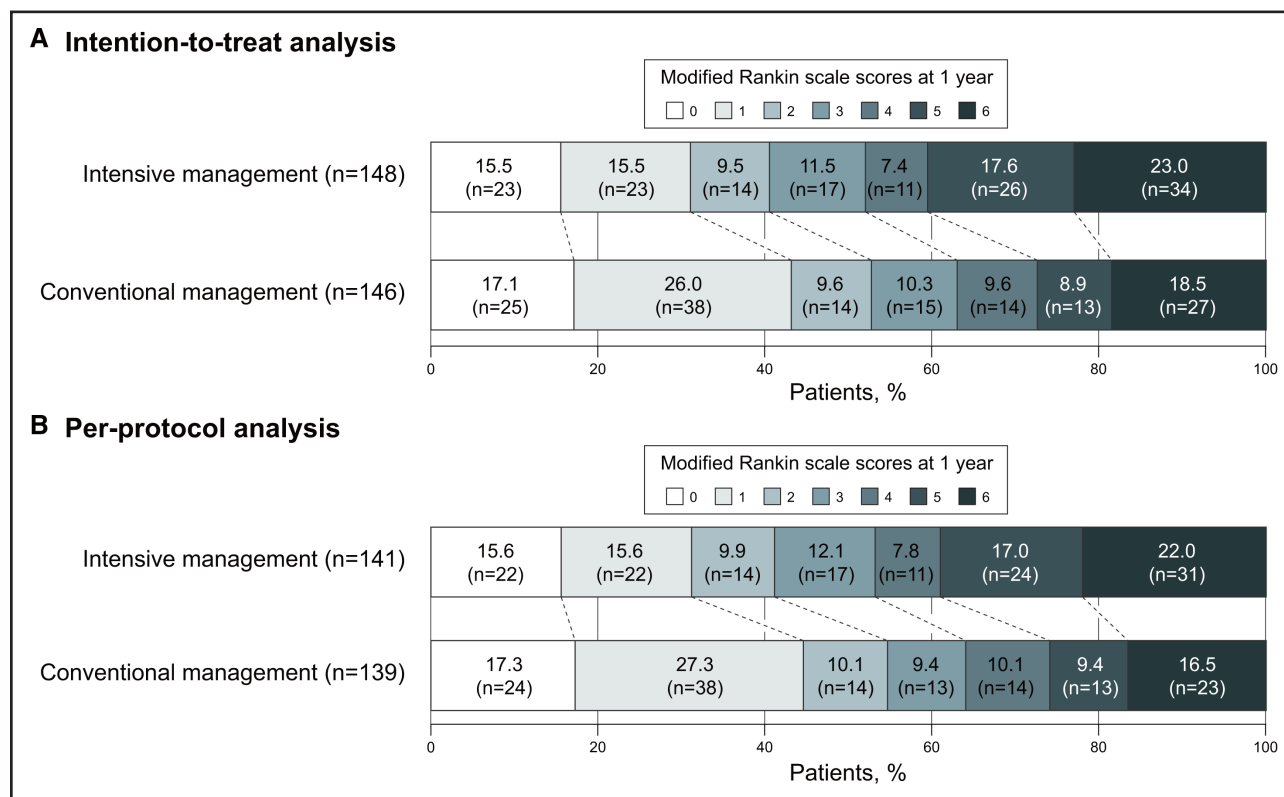
*Adjusted for age, sex, onset to randomization time, and the NIHSS at admission.

in either the intention-to-treat analysis (adjusted hazard ratio, 1.30 [95% CI, 0.89–2.17]; $P=0.307$) or the per-protocol analysis (adjusted hazard ratio, 1.36 [95% CI, 0.79–2.34]; $P=0.272$; Figure 3). In analyses of causes of death, ischemic stroke was the leading cause in both groups (intensive, 20/34 [58.8%] versus conventional, 16/27 [59.3%]), and the proportion of stroke-related

deaths did not differ between the intensive and conventional management groups (Table S2).

Secondary Outcomes

The mRS shift analyses showed the intensive management group had marginally worse scores than the

**Figure 2. Distribution of modified Rankin Scale (mRS) scores at 1 year in the intention-to-treat and per-protocol analyses.**

The distribution of mRS score at 1 year is shown for intensive and conventional blood pressure management in the intention-to-treat (A) and per-protocol analyses (B).

Table 3. Primary and Secondary Outcomes in the Per-Protocol Analysis

	Intensive management (n=141)	Conventional management (n=139)	Risk difference, % (95% CI)	Unadjusted OR (95% CI)	Adjusted OR* (95% CI)	P value
Per-protocol analysis						
Primary outcomes						
Functional independence at 1 y (mRS score of 0–2)	58/141 (41.1)	76/139 (54.7)	–13.5 (–25.1 to –1.9)	0.58 (0.36 to 0.93)	0.56 (0.32 to 0.97)	0.040
Mortality within 1 y	31/141 (22.0)	23/139 (16.5)	5.4 (–3.8 to 14.7)	1.42 (0.78 to 2.61)	1.36 (0.71 to 2.62)	0.353
Secondary outcomes						
mRS score increases at 1 y (shift analysis)	3 (1 to 5)	2 (1 to 5)		1.61 (1.06 to 2.45)	1.54 (1.00 to 2.40)	0.054
Excellent functional outcome at 1 y (mRS score of 0–1)	44/141 (31.2)	62/139 (44.6)	–13.4 (–24.7 to –2.1)	0.56 (0.34 to 0.92)	0.51 (0.28 to 0.93)	0.028
Moderate functional outcome at 1 y (mRS score of 0–3)	75/141 (53.2)	89/139 (64.0)	–10.8 (–22.3 to 0.6)	0.64 (0.39 to 1.03)	0.65 (0.38 to 1.13)	0.132
mRS score of 5–6 at 1 y	55/141 (39.0)	36/139 (25.9)	13.1 (2.3 to 24.0)	1.83 (1.10 to 3.06)	1.85 (1.04 to 3.35)	0.038

mRS indicates modified Rankin Scale; NIHSS, National Institutes of Health Stroke Scale; and OR, odds ratio.

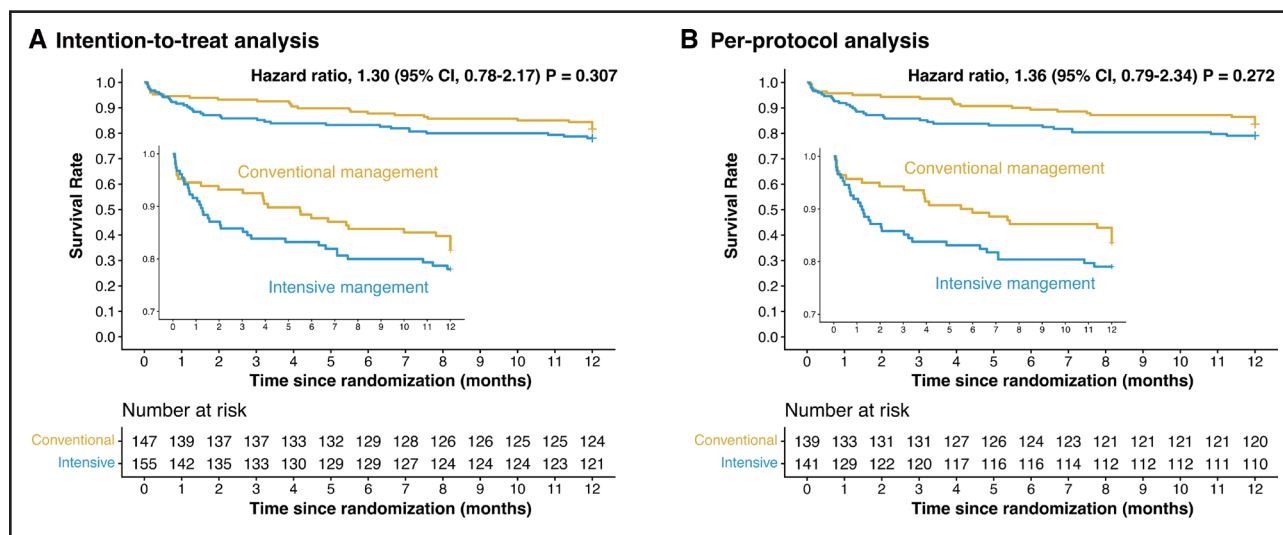
*Adjusted for age, sex, onset to randomization time, and the NIHSS at admission.

conventional management group in either the intention-to-treat analysis (adjusted common OR, 1.52 [95% CI, 0.99–2.35]; $P=0.057$) or the per-protocol analysis (adjusted common OR, 1.54 [95% CI, 1.00–2.40]; $P=0.054$). However, for excellent functional outcome (mRS score, 0–1) at 1 year, the intensive management group showed a significantly lower rate of achievement than the conventional management group in both the intention-to-treat analysis (risk difference, –12.1% [95% CI, –23.0% to –1.1%]; adjusted OR, 0.55 [95% CI, 0.31–0.97]; $P=0.042$) and per-protocol analysis (risk difference, –13.4% [95% CI, –24.7% to –2.1%]; adjusted OR, 0.51 [95% CI, 0.28–0.93]; $P=0.028$). The combined outcome of mRS score of 5 to 6 at 1 year was more frequent in the intensive management group

than in the conventional management group in both the intention-to-treat and per-protocol analyses. For moderate functional outcome (mRS score of 0–3) at 1 year, no significant difference was observed between the 2 groups in either analysis (Table 2; Table 3).

BP Measurements

There were no significant differences in SBP at enrollment between the intensive and conventional management groups (155.3 mmHg [SD, 13.4] versus 154.9 mmHg [SD, 14.4]; $P=0.857$). At 1 hour after randomization, the intensive management group exhibited lower SBP (134.8 mmHg [SD, 19.9] versus 140.7 mmHg [SD, 19.7]; $P=0.008$) than the conventional management

**Figure 3. Kaplan-Meier curves for time-to-death within 1 year in the intention-to-treat and per-protocol analyses.**

Kaplan-Meier curves show overall survival during the 1-year follow-up period in patients assigned to intensive vs conventional blood pressure management after endovascular thrombectomy. **A**, Presents the intention-to-treat analysis and **B** presents the per-protocol analysis. There were no significant differences in overall survival between the 2 groups in either analysis.

group. The 24-hour mean SBP was also lower in the intensive management group (129.4 mmHg [SD, 7.7] versus 138.0 [SD, 13.6]; $P<0.001$). During the first 24 hours after enrollment, intravenous antihypertensive agents were administered in 73.0% of patients in the intensive management group and in 19.2% of patients in the conventional management group ($P<0.001$; Table S3; Figure S1).

Change in mRS Score From 3 Months to 1 Year

In the intensive management group, 24 patients (16.2%) demonstrated an improvement of at least 1 point in the mRS score at 1 year compared with 3 months. In the conventional management group, 18 patients (12.3%) showed such improvement. Conversely, a worsening of at least 1 point in the mRS score between 3 months and 1 year occurred in 32 patients (21.6%) in the intensive management group and in 31 patients (21.2%) in the conventional management group. Overall, there was no significant difference between the 2 groups in the distribution of mRS score changes from 3 months to 1 year ($P=0.609$; Figure 4).

Subgroup Analysis

No statistically significant interactions were identified across any prespecified subgroups for the outcome of functional independence at 1 year. In addition, effect

estimates were consistently in favor of conventional management across all strata (Figure S2).

DISCUSSION

In the 1-year follow-up of the OPTIMAL-BP trial, patients assigned to intensive BP management had a lower likelihood of achieving functional independence at 1 year compared with those assigned to conventional BP management. This difference was statistically significant in the per-protocol analysis, whereas the magnitude of the association was attenuated and reached marginal statistical significance in the intention-to-treat analysis. The likelihood of excellent functional recovery at 1 year was significantly reduced in the intensive BP management group, while no significant difference in 1-year mortality was observed between groups. These results suggest that the adverse functional effects of intensive BP lowering after successful EVT are not transient but may persist into the longer-term recovery phase.

Our findings extend and reinforce the results of prior randomized controlled trials and guideline recommendations that have questioned the benefit of intensive BP lowering after EVT.^{4–7,10} Meta-analyses of post-EVT BP trials have demonstrated harmful effects of intensive BP lowering on functional outcomes at 3 months.^{20–22} Consistent with these observations, current hypertension guidelines recommend against lowering SBP below 140 mmHg during the first 24 to 72 hours after successful

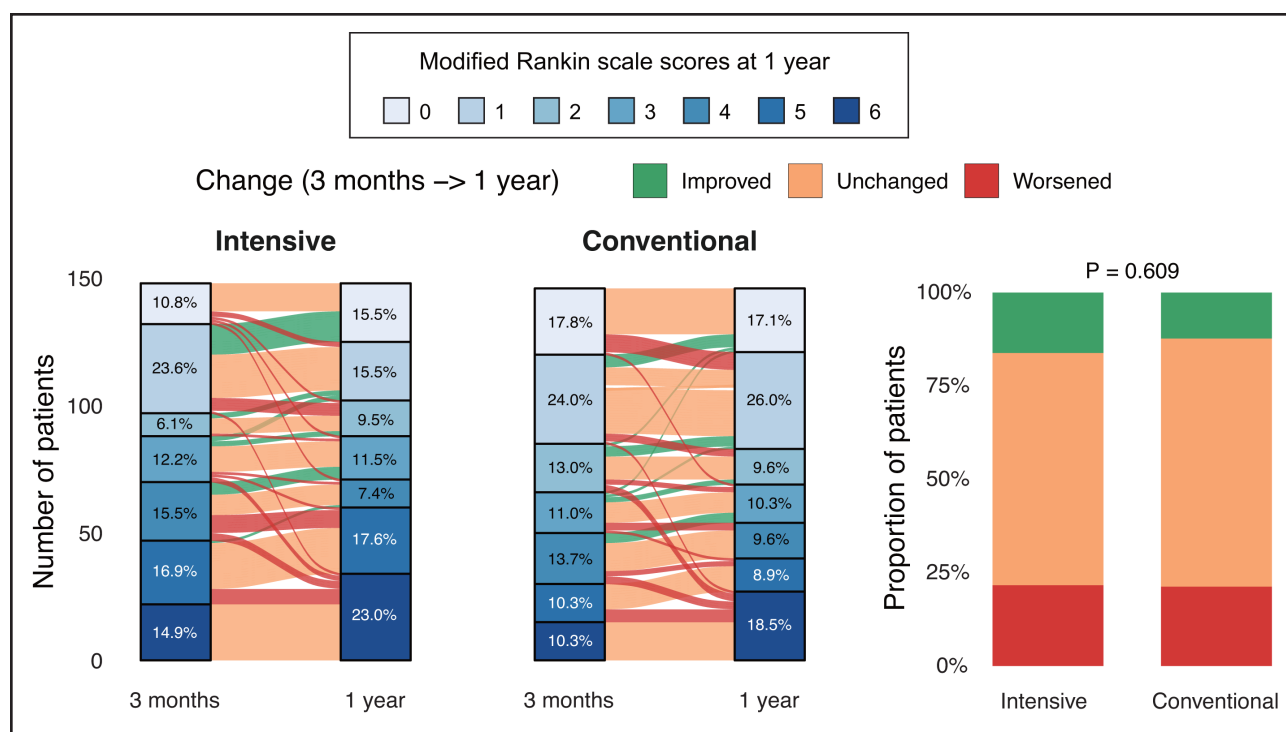


Figure 4. Change in modified Rankin Scale (mRS) score from 3 months to 1 year.

Individual transitions in mRS scores are shown using Sankey diagrams for the intensive and conventional blood pressure management groups. Colors indicate improvement (green), no change (orange), or worsening of disability (red). The proportions of patients with improvement, no change, or worsening did not differ significantly between groups ($P=0.609$).

reperfusion, citing a class III (harm) recommendation.¹⁰ However, whether these 3-month functional differences represented a delay in recovery or a durable harm in long-term outcome remained uncertain. By demonstrating persistent differences at 1 year, our study provides important longitudinal evidence that early BP management strategies may influence not only short-term neurological recovery but also long-term functional trajectories after EVT. To our knowledge, this study is the first to demonstrate that the harmful effects of intensive BP lowering in patients with stroke who achieved successful EVT persist when extended to the 1-year time point.

The observed dissociation between functional outcomes and mortality is particularly noteworthy. Despite lower rates of functional independence and excellent recovery in the intensive management group, we did not observe significant differences in 1-year mortality or time-to-death within 1 year between groups. This pattern implies that intensive BP lowering may primarily affect neurological recovery rather than survival. From a pathophysiological perspective, aggressive BP reduction in the early postreperfusion period may compromise collateral-dependent perfusion,²³ exacerbate reperfusion injury, or impair penumbral tissue recovery, leading to greater infarct growth without necessarily increasing fatal complications.²⁴

Importantly, the absence of meaningful divergence in mRS trajectories between 3 months and 1 year suggests that intensive BP management does not substantially influence the direction of recovery beyond the early poststroke period. The proportions of patients who experienced improvement or worsening of mRS scores during this interval were similar between the 2 groups, and the overall distribution of mRS score changes did not differ significantly. These findings suggest that the adverse effects of intensive BP lowering on functional outcomes likely originate in the hyperacute or early subacute phase after EVT and subsequently persist, rather than emerging progressively during later recovery. In this context, early BP management appears to establish an initial recovery plateau, with limited opportunity for subsequent functional compensation or catch-up over time.

Several limitations warrant consideration. First, the OPTIMAL-BP trial was terminated earlier than initially planned. Early termination may reduce the statistical power of the trial and can increase the uncertainty around effect estimates. In particular, a smaller-than-planned sample size may have contributed to the marginal statistical significance observed in the intention-to-treat analysis for functional independence at 1 year. Therefore, the results should be interpreted with caution. Second, postdischarge care, including rehabilitation intensity, outpatient management, and long-term BP control, was not standardized and was managed according to local practice at each participating center. Thus, we were unable to assess or adjust for their potential impact on long-term

functional outcomes. Given that these factors are known to substantially influence recovery after stroke, their imbalance between groups cannot be excluded and may have confounded the observed associations. Third, missing 1-year mRS data were not imputed, which may introduce bias if loss to follow-up was not completely random, although follow-up completeness was high (96.1%). This reduction in sample size due to follow-up loss may have contributed to the marginal statistical significance observed in the intention-to-treat analysis, thereby limiting the ability to draw definitive conclusions regarding the primary outcomes. Fourth, BP management was protocolized only during the first 24 hours after EVT, which was selected based on the prevailing guideline framework and practical feasibility rather than evidence of superiority over a longer 72-hour strategy. Therefore, our findings should be interpreted as applying to the early post-EVT period, and the potential effect of BP management beyond 24 hours remains uncertain. Fifth, while we adjusted for key clinical covariates, unmeasured confounding related to postdischarge care or rehabilitation intensity cannot be excluded. Sixth, intraprocedural BP and BP management before randomization were not systematically collected, limiting our ability to evaluate the potential influence of intraprocedural hemodynamics on clinical outcomes. Seventh, during the interval between completion of EVT and randomization, BP management was not guided by a trial-specific protocol but was left to local practice and treating physician discretion, which may have introduced variability in early postprocedural hemodynamic management and potentially influenced outcomes. Eighth, baseline infarct core and core-penumbra mismatch data were not systematically collected because imaging protocols and assessment methods varied across participating centers, precluding comparison of these characteristics between the treatment groups. Ninth, clinically significant hypotension was not defined by a prespecified numerical threshold in the protocol, and its management was left to the discretion of each site investigator according to the clinical context, which may have introduced inter-center variability. Finally, as the OPTIMAL-BP trial enrolled patients exclusively from South Korea, caution is warranted when extrapolating these findings to populations with different ethnic or geographic backgrounds.

In summary, among patients with acute ischemic stroke who achieved successful EVT, intensive BP lowering targeting SBP <140 mmHg led to persistently worse long-term functional outcomes at 1 year, including lower rates of functional independence and excellent recovery, compared with conventional BP management. Our findings suggest that early post-EVT BP management strategies may have durable implications for recovery and support current guideline recommendations against intensive BP lowering in this setting.

ARTICLE INFORMATION

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The podcast and transcript are available at <https://www.ahajournals.org/str/podcast>.

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Supplemental Material

Tables S1–S3

Figures S1–S2

CONSORT Checklist

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